Evaluation of Multi-Modal MRI Images for Brain Tumor Segmentation

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***Abstract*—Brain tumors are the most threatening types of tumors and are classified into four grades. To capture the contrast variations of each grade of tumor, currently, the Magnetic Resonance Imaging (MRI) images are considered as the gold standard and are preferred due to their non-invasive nature. However, accurate segmentation of these tumors relies on computational models. Existing models use multimodal MRI images for the segmentation of each grade of a tumor which results in downgraded performance in some tumor regions. In this paper, we propose a new method that incorporates a deep learning-based model called U-Net to address the problem of brain tumor segmentation. The conceptual novelty of our method lies in modifying the U-Net model to make it suitable for multi-grade tumor segmentation task. Additionally, we use this architecture to evaluate individual MRI modalities for the segmentation of brain tumors. The proposed method is evaluated on Brain Tumor Image Segmentation (BRATS) 2015 dataset and has achieved significantly good results. It is pertinent to mention here that by using only T_1 modality, we achieved the highest score for complete sub-tumoral region. On the other hand, a better result in core sub-tumoral has been achieved by using T_2 modality, while the highest score in enhancing sub-tumoral region has been achieved with the help of T_f modality.**

***Index Terms*—Brain tumor, Deep learning, Magnetic Resonance Imaging, Tumor segmentation, U-Net**

I. Introduction

Brain Tumors are graded into High-Grade Gliomas (HGG) and Low-Grade Gliomas (LGG) [1], [2]. Due to the aggressive nature of these tumors, the patient survival duration is a year on average [3]. For treatment, these patients undergo different types of surgery, chemotherapy, and other remedies depending on the tumor [4].

According to the World Health Organization (WHO) classification [2], there are 4 grades of brain tumor namely necrosis, edema, non-enhancing and enhancing tumor. To detect these tumors Magnetic Resonance Imaging (MRI) images are used providing good contrast of the soft tissues. Different MRI modalities such as T_1, T_{1c}, T_2 and T_f images are used together for the detection of brain tumors. Each modality provides different contrast variations for different types of brain tissues. For example, in T_1 MRI modality, fat appears bright and water appears dark, which is good in demonstrating the variation in soft tissues of the brain. Whereas the T_{1c}, is the same as T_1 but with contrast enhancement. In T_2 MRI modality fat appears dark and water appears white, which is good in demonstrating pathology of the brain. In T_f MRI modality water appears bright but is suppressed which allows better detection in the water region.

Traditionally, researchers use multiple MRI imaging modalities together, in computational models, for the segmentation of brain tumors [5]–[7]. These works overlook one most important aspect of using an individual modality for some tumoral regions; this may increase the performance of the model. The rationale behind such a choice is that each modality gives different contrast to different regions of the brain independently. Some contrast variations not captured in one modality might be present in another modality. Therefore, in this paper we evaluate four MRI imaging modalities (T_1, T_{1c}, T_2 and T_f) for three sub-tumoral regions, namely, complete, core and enhancing, to see which modality performs better for which type of tumor region. We operate our method on BRATS 2015 dataset.

The rest of this paper is organized as follows: In Section II, previous related literature is discussed to highlight the current state of the art. The proposed method is presented in Section III. The details on the database, setup, and evaluation of the method are mentioned in Section **??**. Results are presented and discussed in Section IV. Finally, we conclude our work in Section V.

II. Literature Review

Manual identification of brain tumors is usually inaccurate and time-consuming [1]. Due to the high chances of error in manual segmentation, automatic or semi-automatic methods are required to help radiologists e.g., in the diagnosis of tumor, treatment panning etc. [8].

Preliminary works in this field focused primarily on constructing features based on different types of shapes and constraints (e.g., rare chance of occurring on cerebrum) of brain tumors [8], [9]. Some researchers used these features and provided them to classification forest for brain tumor segmentation [9]. Another group of researchers reported high dice scores on one region yet failing in other tumor areas [8]. Many other approaches were designed by constructing hand crafted features to achieve the high dice coefficient for the tumor segmentation of each grade of the tumor but not even a single approach achieves high accuracy in all the grades.

Most recently, in a major advance in 2013, the authors in [10], utilized convolutional neural networks by using small 3x3 size kernels for brain tumor segmentation. The proposed architecture was validated in the BRATS 2013 database by achieving the first place for the complete, core, and enhancing regions in dice similarity coefficient metric with values of 0.88, 0.83, 0.77, respectively.

Another group of researchers in [7], proposed a noval approach for designing a three-dimensional convolutional neural network for Brain Tumor Segmentation. To reduce the memory requirements of three-dimensional convolutions, cropped images were fed into their model. Their best performing model achieved a dice score of 0.71 on the testing set, making their model as state-of-the-art architecture. Many most recent models e.g., [11]–[20], in particular, inspired by the technique in [21], three-dimensional constraints model was added in order to achieve high accuracy.

In [22], the authors presented three novel CNN-based architectures to accomplish the task of brain tumor segmentation. Their method was applied on the BRATS challenge dataset. They reported comprehensive segmentation results using a patch-wise approach, and trained it on a balanced class training set of the tumor and non-tumor patches. They achieved a mean dice score of 0.86.

In [23], the authors developed a very plausible method using 2D convolution based segmentation model which is also known as the U-Net architecture. In their method, they applied a number of data augmentation techniques that resulted in high dice scores. This method reported the highest segmentation results for different grades of brain tumors. Our experimental set up bears a close resemblance to the same U-Net architecture and we modify it to evaluate individual modalities for different grades of tumors.

III. Methodology

To evaluate individual modalities for different grades of tumors, we propose a novel method that utilizes a modified version of U-Net architecture. The overview of the proposed method is presented in Figure 1. The proposed method has four major steps namely, 1) MRI Preprocessing, 2) Trained Weights, 3) U-Net Architecture, and, 4) Tumor Segmentation. The following sub-sections contain a step by step explanation of all the major steps involved in our method:

A. MRI Preprocessing

The input of this step is an axial MRI slice extracted from a 3D MRI image. The raw MRI slices have variable levels of detail (which vary from slice to slice), poor contrast values in a high dynamic range, as well as a variable range of values. The slices need to be preprocessed for removing unwanted artifacts and bringing uniformity in the input data before they can be given as input to the classifier. In the first step, we preprocess the input MRI slices using the following three steps:

1) Detailed slice extraction: This process is performed on the entire BRATS2015 HGG dataset which contains 220 patient 3D images. Each patient's image contains 155, 2D slices. We selected only information rich slices in which a significant part of the brain was present. We used sobel edge detector to select such slices.

2) Contrast Stretching: To improve the poor contrast in a high dynamic range values, we applied contrast stretching on all extracted MRI slices. This operation increased the contrast of each slice to the maximum range. We use the following equation to stretch the contrast of each slice:

$$p_pixel = \frac{(GWM - BWM) * (in_pixel - gmin)}{(gmax - gmin)} + BWN \tag{1}$$

where, *GWM* is the maximum desired intensity level, *BWM* is the minimum desired intensity level, *in_pixel* is the value that needed to be stretched, *gmin* is the slice's minimum intensity, and *gmax* is the maximum image intensity

3) Normalization: Lastly, normalization is performed to scale all the pixel values between 0 and 1. This equalized the impact of the features. We extract feature maps from each raw pixel of a slice during the convolutions of U-Net architecture and thus, by normalizing each image, we get the impact of each pixel in a specific range. This way the classifier can learn the variation of each image easily.

B. Trained Weights

The pre-processed slices from the previous step are used to calculate the trained weights vector. The proposed U-Net architecture is trained on the preprocessed slices along with the ground truth associated with them. The trained weights are then utilized in the model for the detection and segmentation of four types of brain tumors as shown in Figure 1.

C. U-Net Architecture

The preprocessed slices and trained weight vector from the previous steps are passed to this step for detecting if the slices have tumor and to segment it. We modify the the classical U-Net architecture to segment the four types of brain tumors, which is previously reported to be the best choice for similar problems [21].

The proposed architecture (as shown in Figure 2), comprises a process of down and up sampling. The process of down sampling is divided into 5 separate blocks, where each block performs similar steps. Each slice of an input tensor goes through 2 layers of convolutions and 1 layer of max pooling operations at each of the five blocks for the reduction in spatial resolution and an increase in feature map size. This process continues until the sixth block appears where the upsampling performs its operations for the rest of the 5 blocks. Each block in upsampling goes through a transpose convolution where the size of the feature map is reduced with the increment of the spatial resolution. Additionally, 2 layers of convolutions are applied with different variants of filter sizes through the upsampling till the output layer is reached. In the end, a kernel of 1x1 is applied with the filters equal to the number of classes i.e., five (one normal class and four tumor classes).

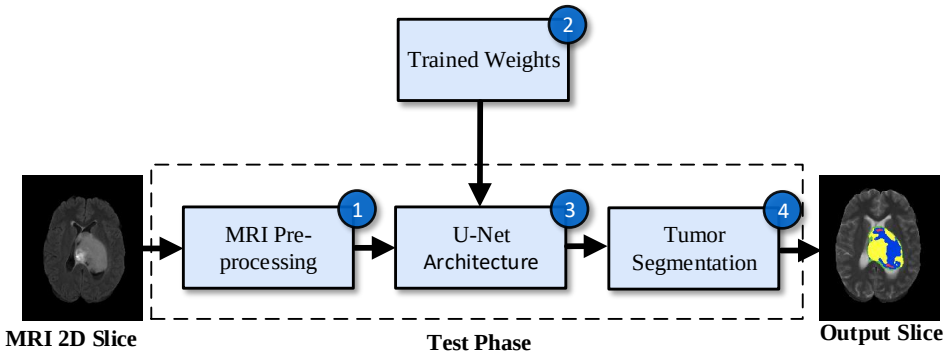

Fig. 1. Representation of step by step processes involved in our method.

In Table I, we present the architecture and configuration of the U-Net model for the detection & segmentation of brain tumors. The table provides a descriptive summary of each layer in the U-Net architecture from start to end. For each phase in the proposed architecture, we utilized different parameters; the choice of parameters along with relevant details are described in the following sections:

1) Activation Function: We employ the Rectifier linear unit (ReLU) [24] activation function in our model. There are other variants of ReLU available in the Keras framework but only the default version works better for our model.

2) Down Sampling: The down sampling in our model is divided into 5 blocks, where each block performs similar repeating steps. A pooling operation is used for down sampling in CNN to reduce the computational load. After many trials, from a number of choices available, we chose the *max pooling* operation for down sampling.

3) Upsampling: In our model, like down sampling, the up sampling is also divided into 5 blocks, where each block applies an augmentation operation to increase the resolution. Among the available choices of the *nearest neighbor, bed of nails, max unpooling*, and *transpose convolution* approaches. We employ the transpose convolution approach [25], for up sampling. For noisy MRI images transpose covolution gives relatively good results.

4) Regularization: In order to generalize the classifier, we use regularization techniques that discard a certain amount of signal to prevent model overfitting. In our model, we use *batch normalization* [26], for regularization.

5) Minimization Function: In the training phase, a loss is to be minimized over the training iterations. We use *adam optimizer* [27], to achieve an accelerated optimization to the error surface. The model passes input tensor through different levels of down and up sampling for creating a segmented map at the output layer. This process is repeated to reduce the error between pixels of ground truth image and the generated output image. When the model's error is converged for all classes, then the resulting trained weights are stored.

TABLE I
THE ARCHITECTURE OF U-NET MODEL FOR THE DETECTION AND SEGMENTATION OF GLIOBLASTOMAS.

Number	Type	Filter size	Stride	Number filters	Input
Layer-1	Conv.	3*3	1*1	64	1*240*240
Layer-2	Conv.	3*3	1*1	64	64*240*240
Layer-3	Max-pool.	2*2	2*2	-	64*240*240
Layer-4	Conv.	3*3	1*1	128	64*120*120
Layer-5	Conv.	3*3	1*1	128	128*120*120
Layer-6	Max-pool.	2*2	2*2	-	128*120*120
Layer-7	Conv.	3*3	1*1	256	128*60*60
Layer-8	Conv.	3*3	1*1	256	256*60*60
Layer-9	Max-pool.	2*2	2*2	-	256*60*60
Layer-10	Conv.	3*3	1*1	512	256*30*30
Layer-11	Conv.	3*3	1*1	512	512*30*30
Layer-12	Max-pool.	2*2	2*2	-	512*30*30
Layer-13	Conv.	3*3	1*1	1024	512*15*15
Layer-14	Conv.	3*3	1*1	1024	1024*15*15
Layer-15	Max-pool.	2*2	2*2	-	1024*15*15
Layer-16	Tranpose Conv.	2*2	2*2	-	1024*30*30
Layer-17	Conv.	3*3	1*1	512	512*30*30
Layer-18	Conv.	3*3	1*1	512	512*30*30
Layer-19	Tranpose Conv.	2*2	2*2	-	512*60*60
Layer-20	Conv.	3*3	1*1	256	256*60*60
Layer-21	Conv.	3*3	1*1	256	256*60*60
Layer-22	Tranpose Conv.	2*2	2*2	-	256*120*120
Layer-23	Conv.	3*3	1*1	128	128*120*120
Layer-24	Conv.	3*3	1*1	128	128*120*120
Layer-25	Tranpose Conv.	2*2	2*2	-	128*240*240
Layer-26	Conv.	3*3	1*1	64	64*240*240
Layer-27	Conv.	3*3	1*1	64	64*240*240
Layer-28	Conv.	3*3	1*1	5	5*240*240

D. Tumor Segmentation

The U-Net architecture, from the previous step, gives a segmented map of the tumor regions, in which each pixel represents one of the five classes. Each pixel value lies in the range of 0 to 4, where, 0 represents normal class, 1 represent necrosis, 2 represents edema, 3 represents non-enhancing and 4 represent enhancing grade tumor. Based on the preceding information, the tumors related to each class are pseudo colored to visually see the segmented regions.

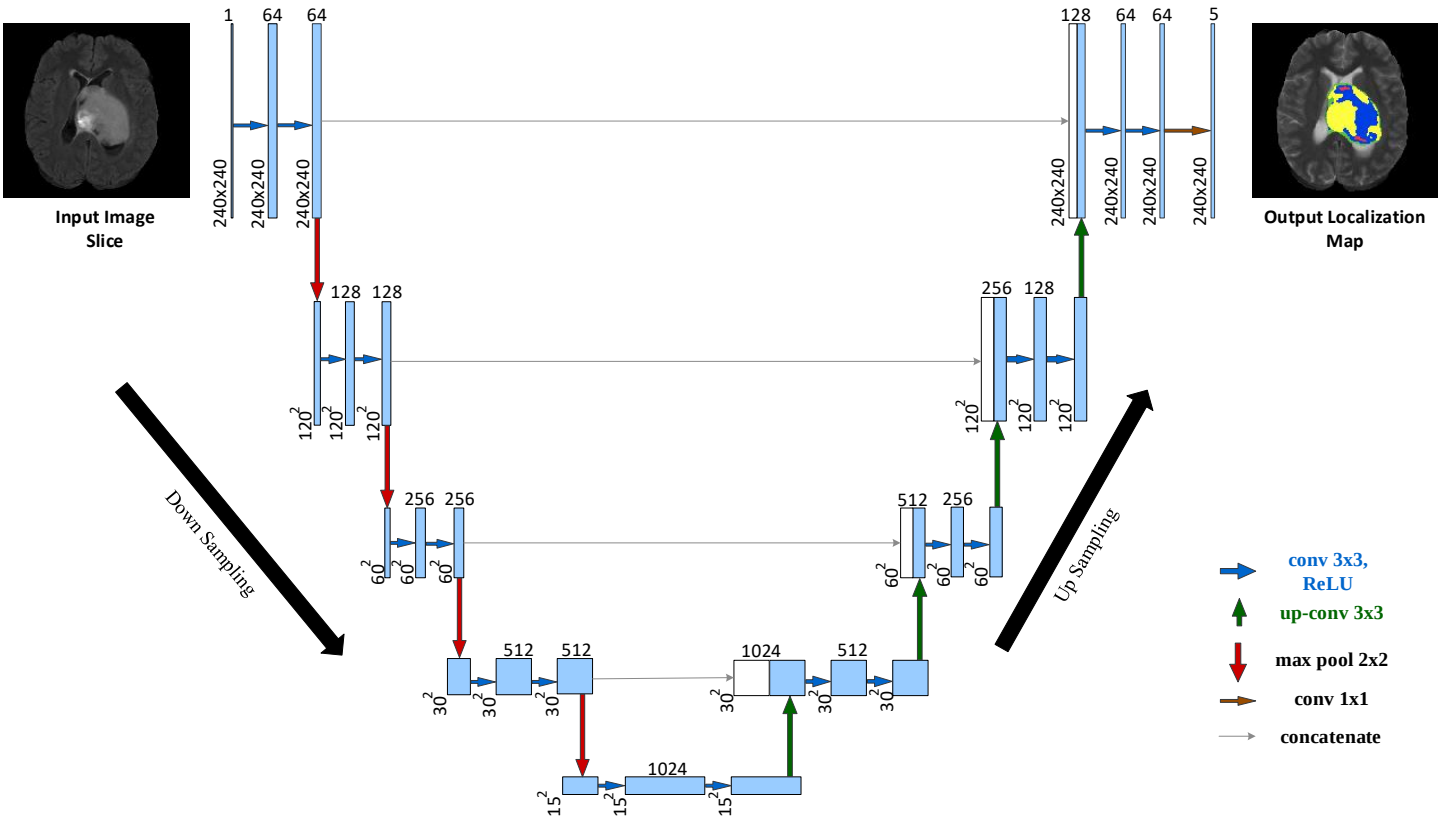

Fig. 2. The overall architecture of the proposed U-Net model, including, convolution, pooling, down & up sampling layers.

IV. Experimental Setup and Results

In this section, we are going to discuss the experimental settings that were used to conduct the experiments along with key results (including efficient segmentation of the tumor regions and ranking the individual modalities). In the following sections we are going to explain each phase in a step by step approach:

A. Dataset

We used BRATS 2015 dataset [8], to carry out our experiments. The dataset contains 4 MRI image modalities, namely, T_1,T_{1c}, T_2 and T_f, along with ground truth. This dataset is divided into HGG (High Grade Glioma) & LGG (Low Grade Glioma) patients. The HGG contains 220 patients and LGG contains 54 patients with 155 no of slices per patient. Each slice resolution is 240 x 240 with a bit depth of 16. We used HGG slices for our experimentation because the nCBF value in the HGG is comparatively higher than LGG, which makes HGG image a better quality image as compared to LGG.

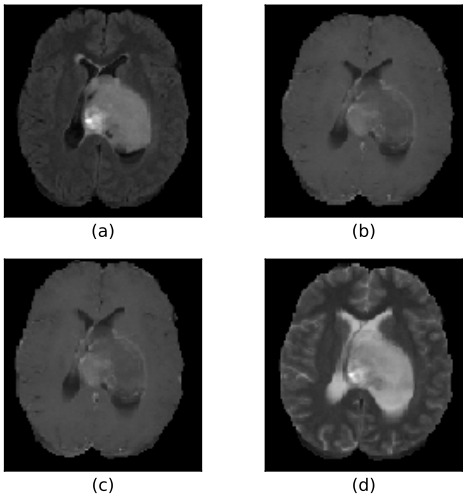

Fig. 3. Four types of MRI images from BRATS 2015 dataset [8], in which, **(a)** T_f modality **(b)** T_1 modality **(c)** T_{1c} modality **(d)** T_2 modality.

B. Setup

To implement the architecture mentioned in Table I, the hyperparameters of the model for the experiments are shown in Table II. These hyperparameters were carefully selected after testing with many variations and the best among all the variations are used. For training, we have used 220 HGG patient 3D images. Images are extracted to individual slices, and preprocessed before training our model.

TABLE II
THE HYPERPARAMETERS USED IN THE PROPOSED MODEL.

Stage	Hyperparameter	Value
Initialization	Weights	Xavier (Default)
Activation Function	ReLU	Alpha=0.0 (Default)
Regularization	Batch Normalization	Mean=0, Std_dev=1
Training	Epochs	80
	Batch Size	12
	Loss Function	Soft Dice Coefficient (Tensorlayer)
	Optimizer	Adam
Model Parameters	Total Parameters	34,536,009
	Trainable Parameters	34,524,231
	Non-Trainable Parameters	11,778

C. Evaluation

For the evaluation of the segmentation done by our model we employ the widely used, Dice Similarity Coefficient (DSC) metric [8]. It provides the overlap measurement between the brain tumoral regions and the segmented outputs. The DSC can be computed using the following formula:

$$DSC = \frac{|P_1 \wedge T_1|}{(|P_1| + |T_1|)/2} \quad (2)$$

where, P_1 is the model prediction and T_1 is ground truth.

D. Sub-tumoral Regions

The tumor area is conventionally grouped into regions [23]. For each patient, three sub-tumoral regions namely, complete, core and enhancing are validated, as shown in Figure 4. The complete region includes all four classes (i.e. necrosis, edema, enhancing and non-enhancing). The core region consists of three classes (i.e., necrosis, enhancing and non-enhancing), while, the enhancing region has only one class (i.e. the enhancing tumor class).

E. Results

The evaluation results using four MRI modalities (individually) for the segmentation of brain tumor for three sub-tumoral regions are shown in Table III. The maximum score for each sub-tumoral region is achieved by a different modalities. For instance, for complete sub-tumoral region, the

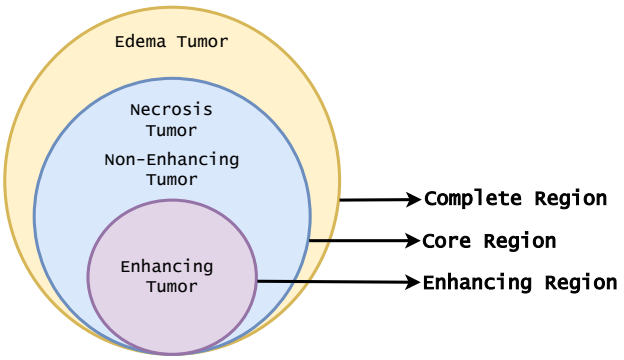

Fig. 4. Pictorial representation of three tumor regions, namely, Complete, Core and Enhancing region. The regions are grouped according to different grades of tumors.

T_1 modality gives the highest score. Whereas, for core sub-tumoral region, T_2 modality gives a higher score than the other modalities, though by a small margin. Lastly, for enhancing sub-tumoral region T_f modality performs better than the other three modalities.

TABLE III
DICE SIMILARITY COEFFICIENT (DSC) OF 3 SUB-TUMORAL REGIONS ON 4 MRI MODALITIES INDIVIDUALLY.

Modalities	Complete	Core	Enhancing
T_1	**0.8841**	0.5177	0.5551
T_{1c}	0.8789	0.5348	0.5459
T_2	0.8829	**0.5356**	0.5426
T_f	0.8689	0.5353	**0.56755**

We also evaluated the segmentation results of all sub-tumoral regions using all four MRI modalities (together). The combined results are shown in Table IV. All four modalities combined outform the individual modalities in core and enhancing sub-tumoral regions. However, for complete sub-tumoral region the T1 modality performs better than all the modalities combined.

TABLE IV
THE DICE SIMILARITY COEFFICIENT (DSC) SCORES OF 3 SUB-TUMORAL REGIONS ON 4 MRI MODALITIES TOGETHER.

Modalities	Complete	Core	Enhancing
T_1, T_{1c}, T_2 and T_f	0.8668	0.5846	0.5775

The main objective of this research was to evaluate individual MRI imaging modalities for brain tumor segmentation. We achieved interesting results by using individual modalities for different sub-tumoral regions. As each modality gives a different type of contrast for different tissue types which makes each modality image unique from other modality. We used this

information as a baseline to segment each grade of tumor using individual modalities and validate the said fact.

V. Conclusion And Future Work

Brain tumors are the most threatening types of tumors and are classified into four grades. Accurate segmentation of these tumors relies on computational models. Existing models use multiple MRI modalities together which results in downgraded performance in some tumor regions. In this work, we evaluated the performance of individual modalities for each sub-tumoral region. We concluded that only for complete tumor region the T1 modality outperforms all the modalities combined. This research can further be extended by introducing residual connections in the current model while reducing regularization. This may improve the performance of the current model.

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